23 **C** is a circle with centre (0, 0)

L is a straight line with negative gradient.

The circle **C** and the line **L** intersect at the points A and B.
The coordinates of A are $(-9, 2)$
The x coordinate of B is 7

Find an equation of **L**.

Circle: $x^2 + y^2 = (-9)^2 + 2^2 = 85$

1 mark
Find circle equation.

At B, $x = 7$: $49 + y^2 = 85$

1 mark
Find y at B.

$y^2 = 36$, so $y = \pm 6$

For negative gradient, B = (7, −6).

1 mark
Choose y = −6.

$$m = \frac{-6 - 2}{7 - (-9)} = \frac{-8}{16} = -\frac{1}{2}$$

Using A: $2 = -\frac{1}{2}(-9) + c$

1 mark
Gradient and intercept.

$c = 2 - 4.5 = -2.5 = -5/2$

$y = -\frac{1}{2}x - 2.5$

1 mark
$y = -\frac{1}{2}x - 5/2$.

(Total for Question 23 is 5 marks)